SQL
Interview Question Bank

Deduplicated, ranked by importance and grouped by topic
Prepared for a Test Automation Engineer with 4+ years of experience

Questions in your original document	**97**
Exact duplicates removed (same question, different wording of punctuation or case)	**0**
Near-duplicates merged (same question asked in different words)	**22**
Unique questions in this PDF	**75**

Section 1: Critical (Must Prepare)	16 questions
Section 2: High Priority	33 questions
Section 3: Medium Priority	20 questions
Section 4: Low Priority / Quick Revision	6 questions

How to Use This Document

Questions are ordered by how likely they are to be asked and how much weight interviewers give them for a tester with 4+ years of experience. Work top to bottom: finish Section 1 before you spend time on Section 3 or 4. Inside each section the questions are grouped by topic, and within each topic the most important question comes first.

A red tag such as **[asked x3 in your bank]** means the same question showed up several times in your original document (in different wording). Repetition is a good signal that it is a frequently asked question.

Section	What it means
Critical (Must Prepare) 16 questions	Asked in almost every SQL round. Be able to write each query by hand, in more than one way where possible, and explain the concept in one or two sentences.
High Priority 33 questions	Frequent follow-ups and practical queries. Practise them after Section 1 using a sample Employee/Orders schema.
Medium Priority 20 questions	Occasionally asked, mostly as a harder follow-up or in SDET-leaning interviews.
Low Priority / Quick Revision 6 questions	Advanced analytical queries that are rarely asked for QA roles. Learn the window-function pattern, not every query.

What Interviewers Look For (4+ Years)

Topic	What the interviewer is testing
Database Concepts & Keys	Clear one-line answers on ACID, keys, normalization and DELETE vs TRUNCATE vs DROP.
Joins & Set Operations	Can you pick the right join, find unmatched rows and explain UNION vs UNION ALL?
Subqueries, Top-N & Ranking	The classic 'second highest salary' family: subquery, LIMIT/TOP, DENSE_RANK, correlated subquery.
Aggregation & GROUP BY	GROUP BY, HAVING vs WHERE, and counts or totals per group, typically written live.
Duplicates & Data Comparison	Finding and deleting duplicates and comparing two tables: very common for QA data validation.
SQL in Testing	How you use SQL as a tester: DB validation, migration, transactions, stored procedures, security.
DML, NULLs, Functions	Everyday query skills: NULL handling, CASE, dates, string functions.
Business Scenario Queries	Turning a business question (latest order, best-selling product) into a correct query.
Advanced Analytics	Window functions and percentile logic; rarely asked for testers, more so for SDET/data roles.

Note: one entry ('SQL Challenge - Queried total balance of employees located in Mumbai') was a resume line; it is now a query task. The 'SQL Query -' prefix was dropped. Repeated questions such as second highest salary, duplicate records and DELETE/TRUNCATE/DROP are merged.

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Topic Finder (all sections for one topic)

Use this if you want to study one topic at a time. Each number links to that topic's questions inside a section.

Topic	S1	S2	S3	S4	Total
Database Concepts & Keys	5 Qs, p.5	2 Qs, p.6	2 Qs, p.8	-	**9**
Joins & Set Operations	2 Qs, p.5	4 Qs, p.6	-	-	**6**
Subqueries, Top-N & Ranking	2 Qs, p.5	7 Qs, p.6	-	-	**9**
Aggregation & GROUP BY Queries	4 Qs, p.5	5 Qs, p.6	2 Qs, p.8	-	**11**
Duplicates & Data Comparison	3 Qs, p.5	1 Qs, p.6	-	-	**4**
SQL in Testing (QA Scenarios)	-	6 Qs, p.6	-	-	**6**
DML, NULLs, Functions & Date/String Handling	-	6 Qs, p.7	10 Qs, p.8	-	**16**
Business Scenario Queries (Orders, Customers, Revenue)	-	2 Qs, p.7	-	-	**2**
Advanced Analytics & Window Functions	-	-	6 Qs, p.8	6 Qs, p.9	**12**

S1 = Critical, S2 = High, S3 = Medium, S4 = Low priority.

Section 1: Critical (Must Prepare) (16 questions)

Asked in almost every SQL round. Be able to write each query by hand, in more than one way where possible, and explain the concept in one or two sentences.

Database Concepts & Keys (5)

1. What are the ACID properties in a database?
2. What is the difference between DELETE, TRUNCATE and DROP? **[asked x2 in your bank]**
3. What is a primary key? Can a table have multiple primary keys?
4. What is a foreign key and why is it important in testing?
5. What is normalization and why is it important?

Joins & Set Operations (2)

6. What are joins and their types? Explain INNER, LEFT, RIGHT, FULL and the difference between inner and outer join. **[asked x3 in your bank]**
7. How do you find records with no matching rows (employees without orders or department, products never sold, rows in one table but not in another)? **[asked x4 in your bank]**

Subqueries, Top-N & Ranking (2)

8. Write a query to find the second highest salary (also: Nth highest). **[asked x3 in your bank]**
9. Retrieve the top N records (top 5 records, top 3 highest paid employees, salaries within the top 3 in the company). **[asked x3 in your bank]**

Aggregation & GROUP BY Queries (4)

10. What is the difference between WHERE and HAVING?
11. Find the total salary and the highest salary in each department. **[asked x2 in your bank]**
12. Show the count of orders per customer and the total revenue per product. **[asked x2 in your bank]**
13. Count how many customers placed more than 5 orders (GROUP BY with HAVING).

Duplicates & Data Comparison (3)

14. How do you find duplicate records in a table? **[asked x2 in your bank]**
15. How do you check for duplicate rows based on multiple columns?
16. How do you delete duplicate rows but keep one instance?

Section 2: High Priority (33 questions)

Frequent follow-ups and practical queries. Practise them after Section 1 using a sample Employee/Orders schema.

Database Concepts & Keys (2)

17. What is an index in SQL?
18. What is a view in SQL and why might QA use it?

Joins & Set Operations (4)

19. What is a self-join and when is it used?
20. How do you join more than two tables?
21. What is the difference between UNION and UNION ALL?
22. Find customers who made purchases but never returned products.

Subqueries, Top-N & Ranking (7)

23. How do you select records with salary higher than the department average?
24. How do you get all records except the highest salary?
25. Write a query to find employees with salary greater than Adam's salary.
26. Find customers with orders above the average order value, or who placed more orders than the average per customer. **[asked x2 in your bank]**
27. What is a correlated subquery?
28. What is the difference between EXISTS and IN?
29. Rank employees by salary within each department (RANK, DENSE_RANK, ROW_NUMBER).

Aggregation & GROUP BY Queries (5)

30. Find the number of employees hired each year.
31. Get the total revenue and number of orders per region, and monthly sales revenue with order count. **[asked x2 in your bank]**
32. Calculate the average order value per customer.
33. Identify top-performing departments by average salary.
34. SQL challenge: query the total balance of employees located in a city (e.g. Mumbai).

Duplicates & Data Comparison (1)

35. How do you check whether two tables have the same data?

SQL in Testing (QA Scenarios) (6)

36. What is SQL and why is it important for a QA engineer?
37. How do you handle transactions in SQL for testing?
38. What are stored procedures and how can QA test them?
39. How do you test SQL injection vulnerability?
40. How do you test data migration when moving from an old DB to a new DB?

41. How do you test data encryption in API/DB?

DML, NULLs, Functions & Date/String Handling (6)

42. How do you check for NULL values and replace them with a default value (IS NULL, COALESCE/IFNULL)? **[asked x2 in your bank]**
43. How do you update multiple columns in a single query?
44. How do you retrieve the last inserted record?
45. How do you fetch records where a column contains a specific substring (LIKE)?
46. How do you use a CASE statement in SQL?
47. How do you get the difference between two dates and find employees hired in the last 30 days? **[asked x2 in your bank]**

Business Scenario Queries (Orders, Customers, Revenue) (2)

48. Get the latest (last) order placed by each customer, along with the order amount; also the first and last order date. **[asked x3 in your bank]**
49. Identify the most selling product.

Section 3: Medium Priority (20 questions)

Occasionally asked, mostly as a harder follow-up or in SDET-leaning interviews.

Database Concepts & Keys (2)

50. What is denormalization and when might QA encounter it?
51. What is the difference between CHAR and VARCHAR?

Aggregation & GROUP BY Queries (2)

52. Find the percentage of employees in each department.
53. Retrieve the maximum salary difference within each department.

DML, NULLs, Functions & Date/String Handling (10)

54. How do you retrieve distinct department names?
55. What is the difference between BETWEEN and IN?
56. How do you retrieve all employees who joined in a given year (2023) or on weekends? **[asked x2 in your bank]**
57. How do you get the count of rows without using COUNT(*)?
58. How do you check if a table has any rows without fetching the data?
59. How do you select random rows from a table?
60. How do you convert a string to uppercase and find its length? **[asked x2 in your bank]**
61. How do you handle special characters in SQL queries?
62. How do you select only numeric values from a column with mixed data?
63. How do you pivot data in SQL?

Advanced Analytics & Window Functions (6)

64. Calculate cumulative revenue by day.
65. Find the moving average of sales over the last 3 days.
66. Show product sales distribution (percent of total revenue).
67. Find churned customers (no orders in the last 6 months).
68. Calculate revenue generated from new customers (first-time orders).
69. Calculate year-over-year growth in revenue.

Section 4: Low Priority / Quick Revision (6 questions)

Advanced analytical queries that are rarely asked for QA roles. Learn the window-function pattern, not every query.

Advanced Analytics & Window Functions (6)

70. Retrieve customers who made consecutive purchases (2 days).
71. Find customers who placed orders every month in 2023.
72. Calculate the average time between two purchases for each customer.
73. Retrieve the longest gap between orders for each customer.
74. Find products that contribute to 80% of the revenue (Pareto principle).
75. Detect customers above their historical 90th percentile purchase amount, or with revenue below the 10th percentile. **[asked x2 in your bank]**